

Problem-Solution Fit

Date	5 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Farmers - Agricultural Researchers - Agri-business Companies - Agricultural Extension Officers - Policy makers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited technical knowledge. - Limited access to expert agricultural advice. - Unpredictable weather conditions. - Financial constraints. - Limited internet connectivity in rural areas.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional farming advice: - Pros: Easy to access, based on local experience. - Cons: Subjective and may not consider current environmental conditions. Manual agricultural research: - Pros: Detailed information. - Cons: Time-consuming and requires expertise. Online farming resources: - Pros: Easily accessible. - Cons: Generic recommendations, not personalized.
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Difficulty selecting the most suitable crop for current soil conditions.
- Low crop productivity due to poor crop selection.
- Limited access to data-driven agricultural recommendations.
- Frequent uncertainty caused by changing environmental conditions.

9. PROBLEM ROOT / CAUSE [RC]

- Lack of accurate, personalized crop recommendations.
- Inadequate analysis of soil nutrients and environmental factors.
- Dependence on traditional decision-making methods.
- Limited access to intelligent agricultural technologies.

7. BEHAVIOR + ITS INTENSITY [BE]

- Frequently consults local farmers or experts before planting.
- Searches online for crop recommendations.
- Depends on previous farming experience.
- Regularly monitors weather and soil conditions.

3. TRIGGERS TO ACT [TR]

- Need to maximize crop yield.
- Changes in soil nutrients or weather conditions.
- Desire to reduce farming risks and financial losses.

10. YOUR SOLUTION [SL]

OptiCrop is a machine learning-based crop recommendation system that analyzes soil nutrients (N, P, K) and environmental parameters (temperature, humidity, pH, and rainfall) to recommend the most suitable crop. The system provides fast, accurate, and data-driven recommendations through an easy-to-use Flask web application, helping farmers improve productivity, optimize resource utilization, and support sustainable agriculture.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Agricultural websites
- Mobile applications
- Government agriculture portals
- YouTube farming channels

4. EMOTIONS BEFORE / AFTER [EM]

Before:

- **Uncertain**
- **Confused**
- **Worried**
- **Hesitant**

After:

- **Confident**
- **Satisfied**
- **Hopeful**
- **Motivated**

OFFLINE

- Agricultural extension officers
- Local farming communities
- Farmer training programs
- Agricultural exhibitions